

Characterizing Space-Time Evolution through Black Hole Merging's Impact on Local Space-Time Fabric

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Background/Abstract

Black hole (BH) merging can reveal how matter behaves in the surrounding space-time. Since our understanding of these events comes from very limited observations, accurate simulations can provide insights into key factors impacting the merging processes and BH structure.



Figure 1: Merging of two black holes (source: universetoday.com)

Analysis of local equilibrium points in the **2-Body merging** scenario can pinpoint reasons as to which characteristics most influence BH during merging by providing descriptions of the surrounding fabric.

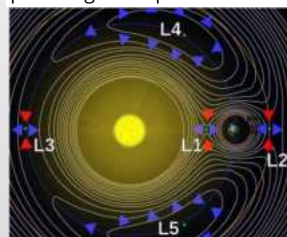


Figure 2: LPs L1-L5 for 2-Body scenario (source: perfectastronomy.com)

Lagrange Point (LP) stability is very useful since it can identify how the surrounding space-time is impacted by the BH merging event. It can be applied to analyze instabilities and fluctuations occurring during these events.

NASA project led by Schnittman (2010) studied the impacts of BH merging on LPs L4 and L5 through simulations. Similar simulations are performed in this project, and they aim to develop better understanding of orbital mechanics and BHs.

Research Question

How do Two Merging Black Holes Effect Local Lagrange Point Stability?

Hypotheses

- When BHs merge, I expect all LPs (L1-L5) to become less stable, with L4 and L5 shifted towards the secondary mass and the rest shifted away from it.
- This is because the emissions released due to the merging will cause distortions in the space-time and thus affect the stability of equilibrium points.
- With the stability changes, those points in line with the two bodies (L1-L3) will go further out due to radiation going straight in their direction, while the adjacent points (L4 and L5) will shift towards the BH.

Methodology

Independent Variables: Accretion Rate, Dynamical Friction, Mass

Dependent Variables: Lagrange Stability, Time

Constants: Black hole masses, Python libraries, device used (laptop), 10 AU Hardness (Accretion Rate Trials), Black Hole Type (Schwarzschild)

Repeated Trials: Universe Sandbox Simulation Temperature Investigations repeated 10 times; custom simulation work repeated 10 times

Materials: AOS Lenovo ThinkPad & Lenovo GEFORCE RTX Legion Laptops

Simulation System Creation

Tools for Creating Simulation System



Figure 3: Python libraries used in this project (source: HTollervay, N.(n.d.). Tutorials. <https://codewith.mu/es/tutorials/1.1/>)

Initial Conditions Set for Simulation

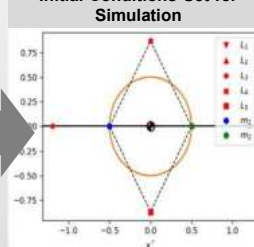


Figure 4: Initial locations and velocities of particles in the simulation and the locations of all analyzed Lagrange Points.

Simulations Created with Animations

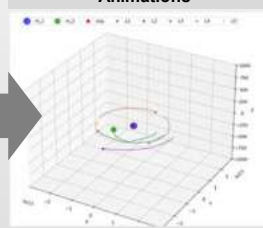


Figure 5: Visual representation of data collected and analyzed in the simulations.

Data Collection and Analysis

Universe Sandbox used to obtain "expected" values/trends given changes to temperature and hardness

Analysis was conducted for these trends and outputs

A new N-Body simulation was created via Python & associated libraries based on the previous simulation and variables

Black Hole characteristics integrated into a Newtonian space-time based model, and LP stability was evaluated as **Relative Displacement**

LP stability analysis per IV to determine trends and impact on surrounding space-time; ML operations conducted to examine any underlying trends

Physics Principles

Angular Velocity
Inertial Reference Frame
Universal Gravitation
Schwarzschild Radius
State Vector
Differentiation with Respect To Time

Results

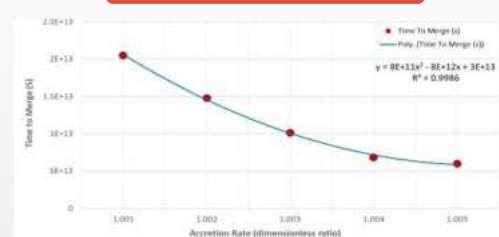


Figure 6: Impact of Accretion Rate (dimensionless) on time merge (s).

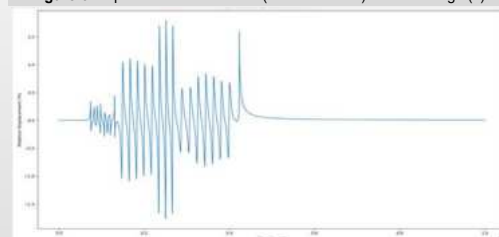


Figure 7: Analyzing the impact of Accretion Rate on LP stability.

Conclusions

- Simulations showed that increasing accretion rates correlate to lower merging times (Figure 6), supporting the hypothesis described which stated the expectation for mass exchanges to occur quicker with greater ensuring accretion rates, likely correlating to lower merging times.
- Hardness initially demonstrated a direct relationship with the BH merge time; however, recent simulations showed an inconsistency in achieving merging, which can be potentially be explained by initial conditions correlating to a gravitational slingshot versus other conditions proceeding in a standard manner
 - These initial conditions were not expected to cause dynamic of this kind; however, for consistency purposes, their influence may need to be placed under consideration for potential inconsistencies in the results achieved
- LP stability in Accretion-adjusted trials have correlated to "gravitational bursts" where LP stability permanently alters in its Relative Displacement and ensues a completely different trajectory
 - This is a topic still under analysis; however, this can correlate to matter-like propulsions which can be explained by large radiation releases in the merging event
 - L4 & L5 have often displayed a sinusoidal-like Relative Displacement for mass-adjusted accretion trials
 - Future work includes a verification publication using a full-relativity model

References

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